

Article

Structural Optimization and SVPWM Control Strategy of Rotary Motors for Plasma Spraying Applications

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Abstract

This study systematically investigates the structural optimization and control strategies of a plasma power supply-based rotating electrical machine. Firstly, stress simulation analysis was conducted on both conventional and optimized motor structures using ANSYS 2025 R1 software. The results demonstrate the maximum stress at the motor bearings decreased from 1.295 MPa to 0.865 MPa after optimization, representing a 33.2% reduction. Secondly, dynamic balance simulation performed with Adams 2024 software revealed that the centroid offset range of the optimized motor was reduced from ± 0.05 mm to ± 0.0175 mm, achieving a 65% improvement. Furthermore, a motor driver board supporting SVPWM and FOC algorithm was designed and implemented, featuring wide voltage input, multiple output channels, and comprehensive protection functions. Experimental verification confirmed that the developed control system could generate ideal three-phase saddle wave and sinusoidal current waveforms, ensuring smooth motor operation. The system demonstrated excellent dyne pen test results on plasma-sprayed acrylic plates, effectively validating the feasibility of both structural optimization and control strategies. The research outcomes provide theoretical foundations and technical support for high-performance motor design in demanding applications such as plasma spraying.

Keywords: rotary motor; stress structure optimization; dynamic balancing; plasma spraying; Space Vector Pulse Width Modulation (SVPWM); Field-Oriented Control (FOC) algorithm

1. Introduction

As a core component for electromechanical energy conversion, the performance of rotating electric machines directly impacts the stability, precision, and energy efficiency of industrial automation equipment, precision machining systems, and specialized applications (e.g., plasma spraying, semiconductor manufacturing). Under high-dynamic, high-load operating conditions such as plasma spraying, these machines must not only endure intense thermomechanical cyclic stresses over prolonged periods but also maintain stable rotational speed and rapid response to ensure coating uniformity and process consistency. However, conventional rotating machines often face two major bottlenecks during high-speed operation: stress concentration and fatigue damage induced by structural design, particularly in critical components such as bearings, which are prone to premature failure due to excessive stress; and vibration and noise caused by dynamic imbalance, which degrade control accuracy and accelerate mechanical wear. Additionally, the control strategy of motor drive systems is a key determinant of dynamic performance and energy efficiency [1]. Space Vector Pulse Width Modulation (SVPWM) combined with



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Field-Oriented Control (FOC) has become the mainstream solution for enhancing AC motor drive performance [2]. By enabling vector decoupling, this approach achieves independent control of torque and flux similar to DC motors, significantly improving dynamic response, operational efficiency, and low-speed stability. Nevertheless, in complex power supply environments with frequent load disturbances—such as plasma power systems—the effective integration of high-reliability hardware design and high-precision control algorithms remains a critical challenge in current engineering practice.

Extensive research has been conducted by domestic and international scholars on the structural optimization of rotating electric machines. Early studies primarily focused on material selection and thermal management design, such as adopting high-strength aluminum alloy housings or optimizing stator slot geometries to reduce core losses. With the increasing maturity of computer-aided engineering (CAE) technologies, finite element analysis (FEA) and multibody dynamics simulations have become indispensable tools for structural optimization. Table 1 lists the currently mainstream structural optimization and control strategies. Yang Weize et al. established a dynamic model of the compressor motor spindle and optimized the counterweight mass using ADAMS software [3]. Wang Z. et al. proposed an optimization strategy for mass compensation in mechanical online dynamic balancing systems, along with establishing a corresponding optimization model [4]. The multi-objective optimization framework based on machine learning and computational fluid dynamics, as proposed in reference [5], can be extended to the methodological foundation of the structural optimization in this paper. Additionally, reference [6,7] systematically investigated innovative cooling structures for batteries and cold plates, providing significant references for heat transfer mechanisms and thermo-structural co-design under complex working conditions. However, most existing studies concentrate on single-physics optimization (e.g., structural mechanics or dynamics), while systematic research on the co-optimization of motor structure and control strategies under complex operating conditions—such as plasma spraying—remains relatively scarce. This research employs Ansys software for stress analysis of the motor and utilizes Adams for simulating the center-of-mass deviation of the motor, achieving coordinated optimization across multiple physical fields (e.g., structural mechanics and dynamics). In response to the nonlinear load characteristics of the plasma process, a Field-Oriented Control (FOC)-based SVPWM algorithm is adopted for motor drive control.

Table 1. The currently mainstream structural optimization and control strategies.

Optimization Design Types	Methods
dynamic balancing design	stress optimization of balance block using ADAMS software [3].
centroid optimization design	establishment of a mass compensation optimization model [4].
motor control algorithms	adoption of model predictive control (MPC) or sliding mode variable structure control [8].

In the field of motor drive and control strategies, Space Vector Pulse Width Modulation (SVPWM) and Field-Oriented Control (FOC) technologies have established a relatively comprehensive theoretical framework since their introduction in the 1990s. Domestic research teams, such as those from Huazhong University of Science and Technology and Zhejiang University, have made significant advancements in SVPWM algorithm optimization, dead-time compensation, and sensorless control techniques. Internationally, companies like Texas Instruments and STMicroelectronics have developed dedicated motor control chips with integrated FOC algorithms, substantially lowering the barrier to implementation. In recent years, advanced control algorithms such as Model Predictive Control (MPC) and sliding mode variable structure control have been progressively introduced to the field of motor control to address challenges posed by nonlinear loads and

parameter disturbances [8]. However, in application environments characterized by strong electromagnetic interference and wide voltage fluctuations, such as plasma power supplies, critical issues including interference-resistant drive board design, multilevel inverter topologies, and online parameter identification still require in-depth investigation. Particularly, while pursuing high-precision current control, ensuring robust hardware design with safety features such as short-circuit protection and undervoltage lockout presents additional engineering challenges.

To address the aforementioned challenges, this study focuses on rotating electric machines in plasma spraying equipment. The research methodology comprises four key aspects: first, Ansys-based static structural simulations are conducted to comparatively analyze stress distributions in critical components (e.g., bearings) between conventional and optimized motor designs. Second, multibody dynamics simulations using Adams are performed to quantitatively evaluate improvements in mass-center deviation and vibration characteristics. Building upon these analyses, a motor drive board with multi-protection functionality is developed, employing an AT32-series MCU as the main controller to implement FOC-based SVPWM algorithms. Finally, experimental validation is carried out through comprehensive testing of drive waveforms, phase current responses, and operational performance, with additional plasma spraying experiments demonstrating the system's enhanced performance under real-world operating conditions.

Current research on rotating electric machines under complex operating conditions such as plasma spraying predominantly focuses on single-physics optimization (e.g., structural mechanics or dynamics), lacking systematic co-analysis of the synergistic relationship between mechanical design and control strategies. To address this gap, this study establishes a closed-loop methodology framework integrating “structural stress/dynamic balance co-simulation with motor control strategy validation,” enabling design optimization and experimental verification of structural improvements combined with SVPWM control strategies. Section 2 presents a detailed Ansys-based stress simulation analysis, including comparative structural analysis between conventional and optimized motor designs [9], stress simulation of conventional motor configurations, and stress evaluation of the optimized motor structure. Section 3 elaborates on Adams-based dynamic balance simulations, encompassing dynamic balance analysis of conventional motors, optimized motor dynamic performance assessment, and thermal–physical field simulation results interpretation. Section 4 details the motor drive board's hardware–software co-design, featuring hardware architecture development for the drive board, and implementation workflow of the FOC algorithm. Section 5 systematically evaluates motor performance through dynamic balance structural testing, drive board output waveform analysis, and plasma-loaded spraying performance validation. Finally, Section 6 discusses the potential applications and engineering value of the optimized motor system.

2. Ansys Stress Simulation Analysis of Rotary Motor

2.1. Comparative Analysis of Structural Design Between Conventional and Optimized Rotary Motors

Figure 1a presents the SOLIDWORKS 2024 structural schematic of the conventional rotating electric machine, while Figure 1b illustrates the optimized design. Table 2 provides a comparative analysis of the structural designs before and after optimization [10–13]. The optimized motor employs a single-bearing support structure, which not only reduces material costs but also simplifies manufacturing processes compared to the original dual-bearing configuration. Furthermore, the optimized design utilizes an internal slip ring (brass material) as the motor anode, replacing the conventional approach of using the housing (aluminum alloy material) as the anode. This modification enhances electrical conductivity while eliminating housing electrification, thereby significantly improving

operational safety. In addition, when employing a single-bearing support structure, to ensure shaft stiffness under high-speed rotation and suppress cantilever deflection, the optimized length-to-diameter ratio (L/D) of the motor shaft has been controlled within ≤ 5 through calculation, a range widely recognized in engineering practice as a reasonable threshold for ensuring shaft rigidity and avoiding bending resonance.

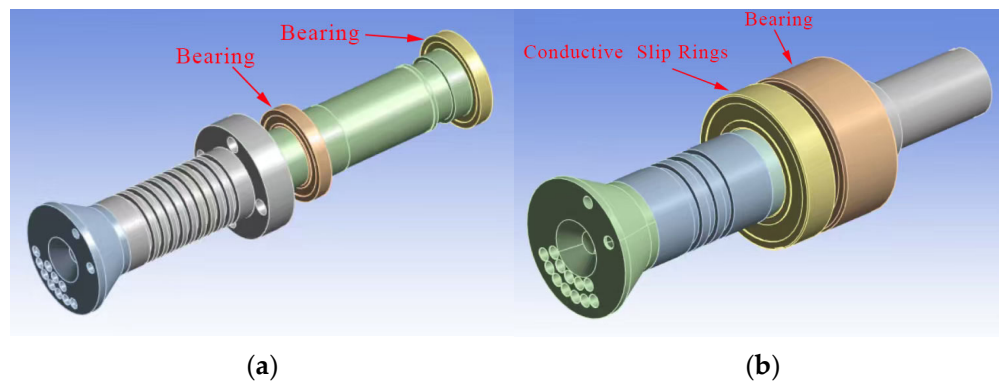


Figure 1. Structural schematics of conventional vs. optimized rotary motors in SolidWorks. (a) Conventional rotary motor structure schematic in SolidWorks; (b) optimized rotary motor structure schematic in SolidWorks.

Table 2. Comparison of structural designs between conventional and optimized rotary motors in SolidWorks.

Type of Rotary Motor	Bearing Design	Anode Conduction
conventional rotary motor	dual-bearing	housing
optimized rotary motor	single-bearing	internal slip ring

Regarding the Ansys stress simulation parameter settings for a rotary motor, Table 3 details the boundary conditions, mesh density, material properties, and optimization criteria used in the ANSYS simulation. Additionally, the allowable centroid offset is within a range of ± 0.02 mm.

Table 3. The Ansys stress simulation parameter settings for a rotary motor.

Parameters	Conventional Rotary Motor	Optimized Rotary Motor
boundary condition	after applying a 3 s torque to the rotating shaft, a speed of 2250 r/min is achieved	
mesh density	the minimum edge length is 9.4248 mm	
material properties	aluminum alloy 6061 with the stress of 395.8 MPa	aluminum alloy 7075 with the stress of 693 MPa
optimization criteria	the maximum stress in the optimized structure is 0.9 MPa	

2.2. Ansys Stress Simulation Analysis of Conventional Rotary Motor

Figure 2 displays the Ansys stress simulation results of the conventional rotating electric machine. The simulation reveals significant stress concentration at the tail bearing, with a maximum stress value of 1.295 MPa [14].

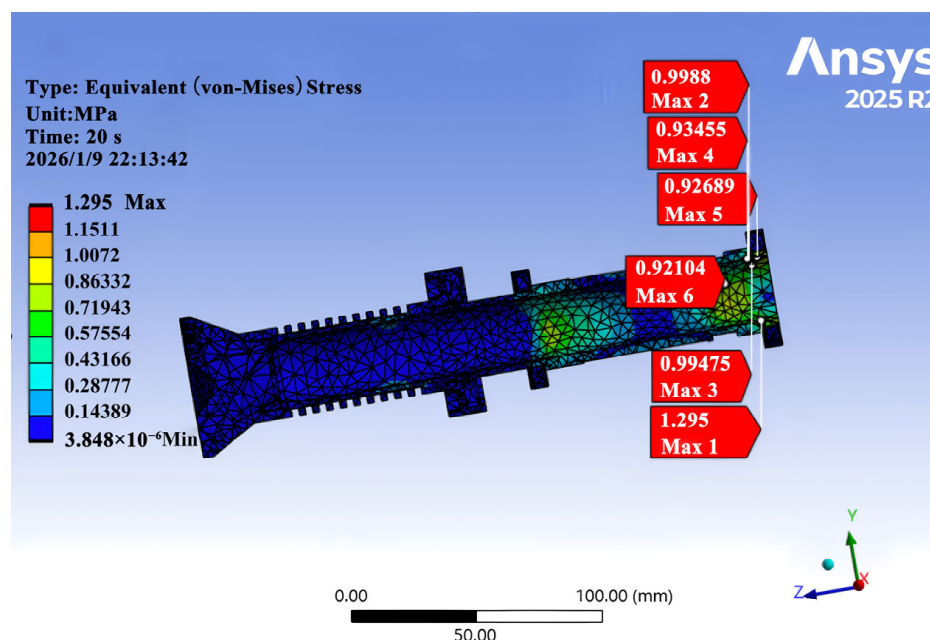


Figure 2. Ansys stress simulation results of conventional rotary motor.

2.3. Ansys Stress Simulation Analysis of Optimized Rotary Motor

Figure 3 presents the Ansys stress simulation results of the optimized rotating electric machine. The simulation demonstrates that stress concentration occurs at the intermediate bearing, with a maximum stress value of 0.865 MPa. Compared with the maximum stress shown in Figure 2, this represents a 33.2% reduction in stress magnitude.

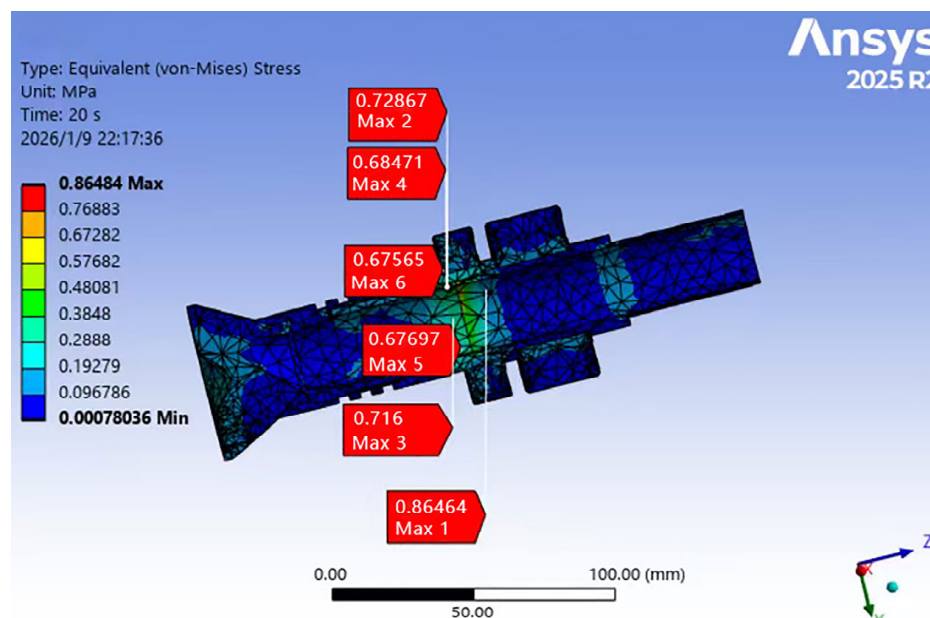


Figure 3. Ansys stress simulation results of optimized rotary motor.

3. Adams Dynamic Balance Simulation Analysis of Rotary Motor

3.1. Adams Dynamic Balance Simulation Analysis of Conventional Rotary Motor

Figure 4 presents the Adams dynamic balance simulation of the conventional rotating electric machine. The simulation results in Figure 5a demonstrate that the centroid offset in the X-axis ranges from -0.05 mm to 0.05 mm, while Figure 5b shows a similar centroid offset range of -0.05 mm to 0.05 mm in the Y-axis.

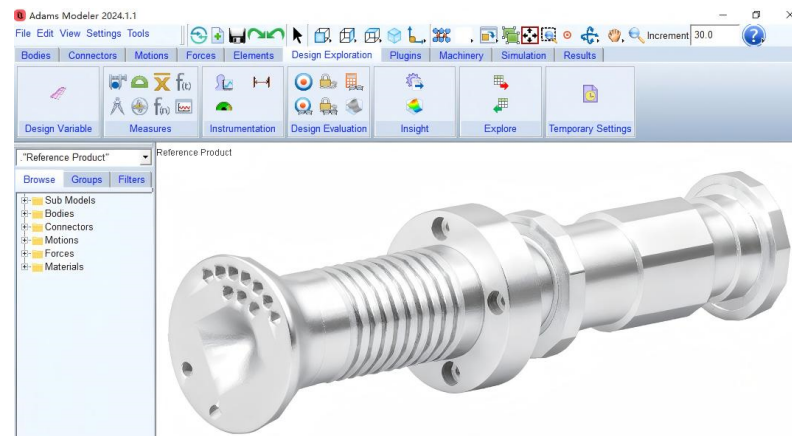


Figure 4. Adams dynamic balance simulation results of conventional rotary motor.

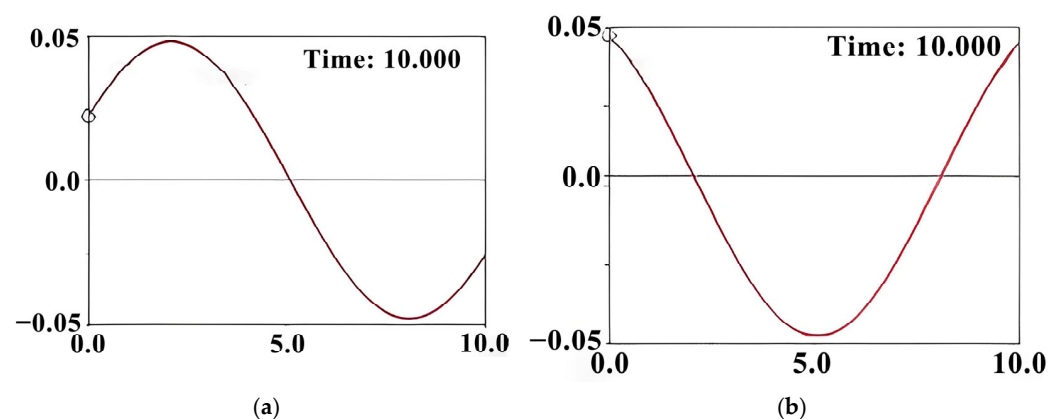


Figure 5. Adams dynamic balance simulation analysis of conventional rotary motor (a speed of 2250 r/min was achieved after applying torque to the shaft for 3 s). (a) X-axis centroid offset of conventional rotary motor; (b) Y-axis centroid offset of conventional rotary motor.

3.2. Adams Dynamic Balance Simulation Analysis of Optimized Rotary Motor

Figure 6 illustrates the Adams dynamic balance simulation results for the optimized rotating electric machine. As shown in Figure 7a, the centroid offset along the X-axis ranges from -0.0175 mm to 0.0175 mm, while Figure 7b demonstrates an equivalent offset range of -0.0175 mm to 0.0175 mm along the Y-axis. Compared with the centroid offset observed in Figure 4 for the conventional rotating machine, these results represent a 65% reduction in vibration amplitude.

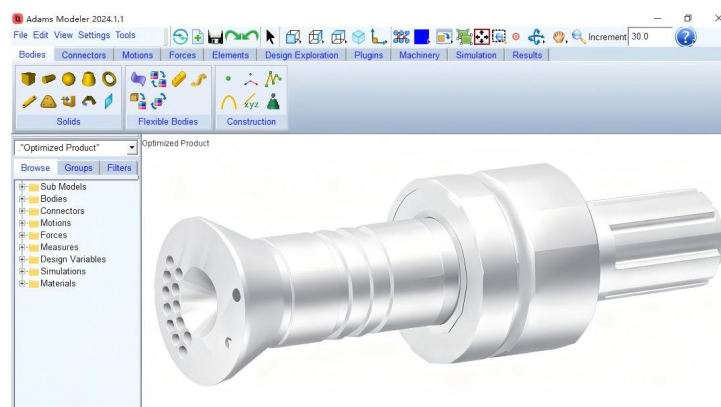


Figure 6. Adams dynamic balance simulation results of optimized rotary motor.

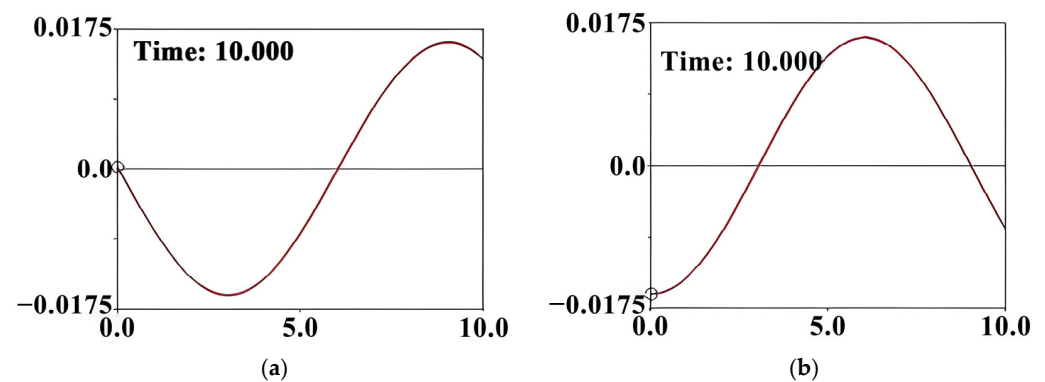


Figure 7. Adams dynamic balance simulation analysis of optimized rotary motor (a rotational speed of 2250 r/min was achieved after applying torque to the shaft for 3 s). (a) X-axis centroid offset of optimized rotary motor; (b) Y-axis centroid offset of optimized rotary motor.

4. Hardware–Software Co-Design of Motor Drive Board

The motor drive board developed in this study incorporates multiple protection features including short-circuit protection, undervoltage protection, and active clamping functionality. Given the strong electromagnetic interference characteristics during plasma power supply operation, the following measures have been implemented at both the hardware and software levels to ensure the robustness of the drive board under complex operating conditions:

(1) Hardware Anti-interference Design:

1. Input Filtering and Isolation: Control signals (PWM) and current sampling signals are electrically isolated using optocouplers or Hall sensors, achieving separation between power ground and control ground and effectively blocking common-mode noise.
2. PCB Design and Grounding: A four-layer board design is adopted, strictly separating the power, control, and ground layers. A “star-point single-point grounding” strategy is implemented to minimize ground-loop interference.

(2) Signal Processing Fault Tolerance and Protection Mechanisms:

1. Signal Processing: Digital filtering is applied to the sampled current and voltage signals. The PI controller in the FOC algorithm incorporates output limiting and anti-windup functions.
2. Fast Protection: Integrated hardware protection circuits (e.g., short-circuit, under-voltage) can respond within microseconds upon fault detection and trigger an MCU interrupt to execute a safe shutdown routine.

4.1. Hardware Architecture of Motor Drive System

Figure 8 presents the hardware system block diagram of the motor drive board, featuring the AT32F403 microcontroller unit (MCU) as the main controller. The system operates with a 24 V DC input to the isolated power module LMR14030SDDAR, which generates 12 V and 5 V outputs. The 12 V output supplies the gate drive circuit, while the 5 V output is further regulated to 3.3 V through the AMS1117 buck converter for MCU power supply. The master AT32 controller generates SVPWM control signals that are transmitted to the gate drive circuit, which subsequently produces SVPWM drive signals for the three-phase half-bridge inverter [15]. The inverter outputs three-phase saddle-wave voltage signals to drive the motor rotation, while the AT32 controller simultaneously performs real-time acquisition of the motor phase currents.

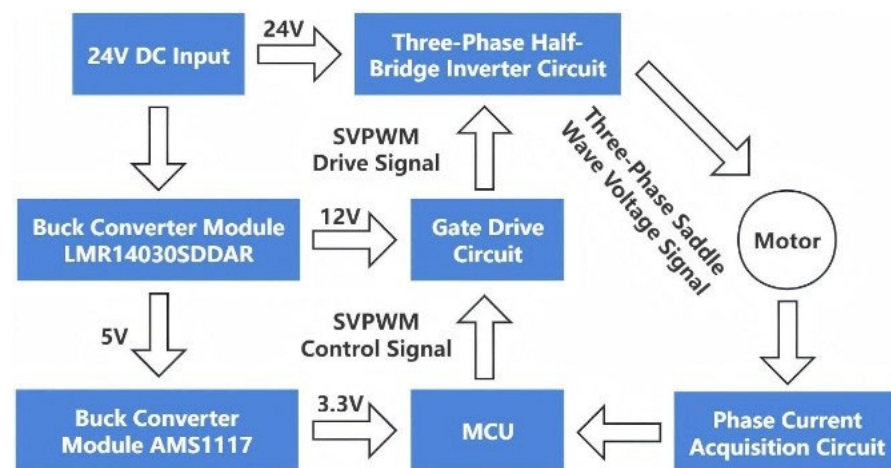


Figure 8. System block diagram of motor drive board hardware.

Figure 9 illustrates the hardware design of the motor driver board, which primarily consists of five functional modules: the microcontroller unit (MCU) module, the voltage step-down module, the gate drive circuit module, the three-phase half-bridge inverter module, and the phase current acquisition circuit module.

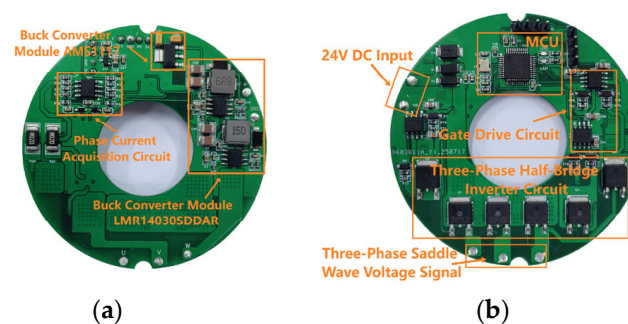


Figure 9. The hardware design of the motor driver board. (a) Front view of the hardware design; (b) rear view of the hardware design.

4.2. FOC Algorithm Design for Motor Drive Board

Figure 10 illustrates the Field-Oriented Control (FOC) algorithm flow implemented in the motor drive board [16]. The main control chip first acquires the three-phase motor currents (I_a , I_b , I_c), which are then transformed into stationary reference frame components (I_α , I_β) via Clarke transformation [17,18]. Subsequently, these components are converted to rotating reference frame quantities (I_q , I_d) through Park transformation [19]. The algorithm compares the obtained I_q and I_d with their reference values (I_{qref} , I_{dref}) to generate error signals, which are processed by PI controllers to produce control voltages U_q and U_d . These control voltages undergo inverse Park transformation to yield U_α and U_β components in the stationary frame. Finally, the system synthesizes the space voltage vector from U_α and U_β , performs SVPWM, and drives the MOSFET switches to control motor rotation. The selection of Proportional-Integral (PI) Controller Gains follows the principle of “tuning the current loop first, followed by the speed loop.” Regarding the selection of PI Controller Gains for the current loop, (1) tuning the Proportional Gain (K_{P_i}): firstly, set the integral gain K_{I_i} to zero. Secondly, begin with a small value of K_{P_i} and increase it gradually until the current response is as fast as possible without introducing high-frequency oscillations. (2) Tuning the Integral Gain (K_{I_i}): with the tuned K_{P_i} fixed, start from a small value of K_{I_i} and increase it until no significant overshoot or low-frequency oscillation is introduced. The current loop is considered properly tuned when the system exhibits a fast step response to current commands, minimal overshoot, and no sustained oscillation.

Regarding the selection of PI Controller Gains for the speed loop, (1) tuning the Proportional Gain (K_{P_s}): firstly, set the integral gain K_{I_s} to zero. Secondly, increase K_{P_s} from a small value until a relatively fast speed response is achieved, while avoiding overshoot or oscillation. (2) Tuning the Integral Gain (K_{I_s}): with the tuned K_{P_s} fixed, increase K_{I_s} from a small value to eliminate steady-state error. The speed should recover quickly with minimal fluctuation following a sudden application or removal of load. The speed loop is considered properly tuned when the speed response is smooth, exhibits no overshoot, and has zero steady-state error. In addition, the following aspects must be considered in the control strategy: the PWM frequency is adjustable within the range of 15–25 kHz; the switching scheme employs Space Vector Pulse Width Modulation (SVPWM); dead-time compensation of the controller is set to 100 ns; current sensor delay is constrained to within 10 μ s; and a rigid rotor simplification is adopted, where in the rotor consists of a shaft with four pairs of oppositely polarized magnets attached in parallel on its surface, neglecting any mass-center offset caused by uneven magnet arrangement.

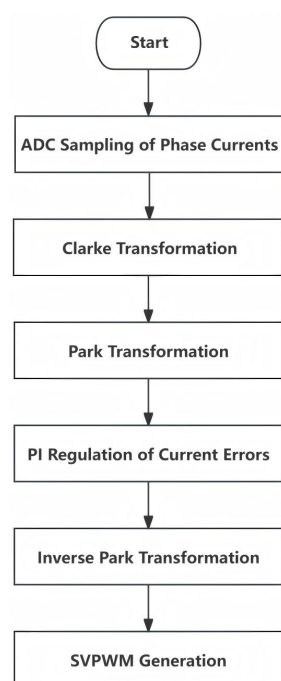


Figure 10. Flowchart of FOC algorithm for motor drive board.

Figure 11 presents the three-phase half-bridge inverter topology employing Space Vector Pulse Width Modulation (SVPWM) [20–23]. This configuration utilizes six power switches to synthesize voltage vectors in a specific sequence, thereby driving the brushless DC motor.

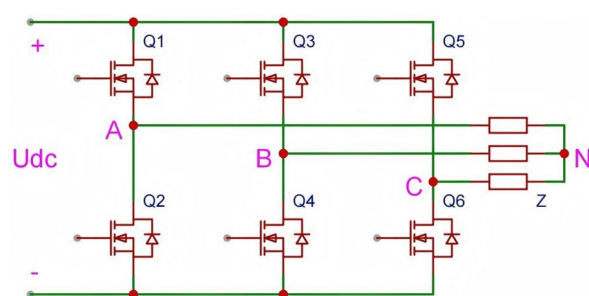


Figure 11. Three-phase half-bridge inverter circuit topology with SVPWM.

Let U_{dc} denote the DC bus voltage of the three-phase half-bridge inverter circuit, with U_{AN} , U_{BN} , and U_{CN} representing the phase voltages of each respective leg [24]. The three-phase half-bridge inverter comprises three half-bridge circuits, where S_x ($x = a, b, c$) represents the switching state of each half-bridge: $S_x = 1$ indicates the upper switch is conducting, while $S_x = 0$ denotes conduction of the lower switch. The switching function S_x is formally defined as follows:

$$S_x = \begin{cases} 1 & \text{(Upper arm conduction)} \\ 0 & \text{(Lower arm conduction)} \end{cases} \quad x = (a, b, c) \quad (1)$$

According to Kirchhoff's voltage law, the relationship between the load phase voltages (U_{AN} , U_{BN} , U_{CN}) and the switching functions S_x can be derived as follows:

$$\begin{cases} U_{AN} = \frac{U_{dc}}{3}(2S_a - S_b - S_c) \\ U_{BN} = \frac{U_{dc}}{3}(2S_b - S_a - S_c) \\ U_{CN} = \frac{U_{dc}}{3}(2S_c - S_a - S_b) \end{cases} \quad (2)$$

Substituting the eight switching states of S_x into Equation (2) yields the phase voltages as listed in Table 4. The three-phase output combinations of S_a , S_b , and S_c comprise eight states, including two zero vectors (000 and 111) and six non-zero vectors.

Table 4. Correspondence between switching states and output voltages.

S_a	S_b	S_c	Vector	U_{AN}	U_{BN}	U_{CN}
0	0	0	U_0	0	0	0
0	0	1	U_1	$-U_{dc}/3$	$-U_{dc}/3$	$2U_{dc}/3$
0	1	0	U_2	$-U_{dc}/3$	$2U_{dc}/3$	$-U_{dc}/3$
0	1	1	U_3	$-2U_{dc}/3$	$U_{dc}/3$	$U_{dc}/3$
1	0	0	U_4	$2U_{dc}/3$	$-U_{dc}/3$	$-U_{dc}/3$
1	0	1	U_5	$U_{dc}/3$	$-2U_{dc}/3$	$U_{dc}/3$
1	1	0	U_6	$U_{dc}/3$	$U_{dc}/3$	$-2U_{dc}/3$
1	1	1	U_7	0	0	0

Figure 12 maps the eight space voltage vector combinations from Table 5 onto the complex plane. The arrow paths in Figure 12 represent the fundamental space voltage vector directions, which divide the complex plane into six sectors. U_{ref} denotes the space voltage vector rotating through each sector during the sampling period, synthesized by the two adjacent fundamental space voltage vectors of the corresponding sector.

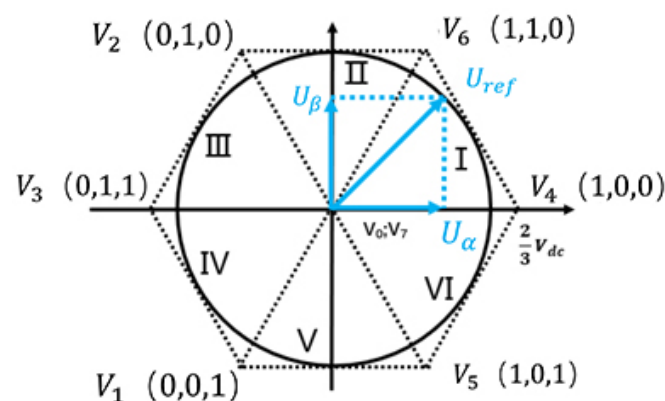


Figure 12. Space voltage vector sectors.

Table 5. Switching patterns for vector synthesis in different sectors.

The Position of U_{ref}	Switching Sequence
I ($0^\circ \leq \theta \leq 60^\circ$)	0-4-6-7-7-6-4-0
II ($60^\circ \leq \theta \leq 120^\circ$)	0-2-6-7-7-6-2-0
III ($120^\circ \leq \theta \leq 180^\circ$)	0-2-3-7-7-3-2-0
IV ($180^\circ \leq \theta \leq 240^\circ$)	0-1-3-7-7-3-1-0
V ($240^\circ \leq \theta \leq 300^\circ$)	0-1-5-7-7-5-1-0
VI ($300^\circ \leq \theta \leq 360^\circ$)	0-4-5-7-7-5-4-0

As shown in Equation (3),

$$\int_0^T U_{ref} dt = \int_0^{T_x} U_x dt + \int_{T_x}^{T_x+T_y} U_y dt + \int_{T_x+T_y}^T U_0 dt \quad (3)$$

This can also be expressed as

$$U_{ref} \cdot T = U_x \cdot T_x + U_y \cdot T_y + U_0 \cdot T_0 \quad (4)$$

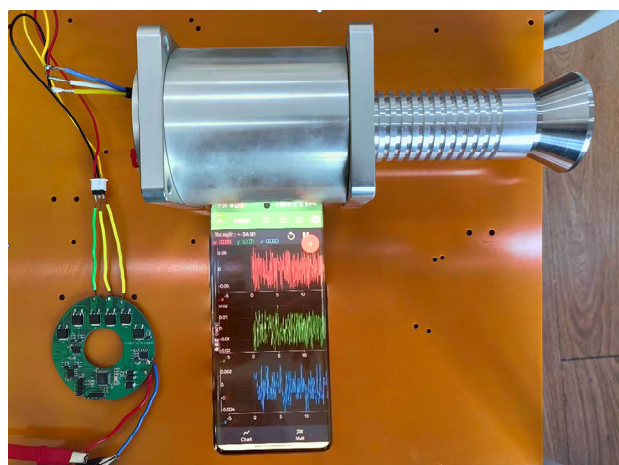
The desired output voltage U_{ref} is synthesized by controlling the dwell times of two adjacent vectors based on the time equivalent principle, with the remaining period compensated by zero vectors. Here, T denotes the PWM carrier sampling period. This synthesis method enables the space voltage vector to incrementally advance by a specific angle each cycle. Since any vector can be synthesized through two constituent vectors, the voltage vector can thus rotate in space, achieving space vector pulse width modulation.

The space vector approach for generating SVPWM waveforms operates as follows: During SVPWM, the control system produces the required reference voltage vector [25] U_{ref} that rotates in space at a specified angular frequency ω . When U_{ref} enters a particular sector of the vector diagram, the system calculates the corresponding space voltage vectors for that sector. The switching patterns for vector synthesis in each sector are summarized in Table 5 [26], where “4” denotes the non-zero vector U_4 (100), indicating the upper switch of phase A (S_a) is turned ON. “6” represents the non-zero vector U_6 (110), corresponding to the upper switches of both phases A and B (S_a and S_b) being activated.

5. Performance Evaluation of Rotary Motor System

5.1. Dynamic Balance Performance Testing

As illustrated in Figure 13, the motor drive board implementing the FOC (Field-Oriented Control) algorithm is connected to the optimized rotary motor.

**Figure 13.** Dynamic balance test setup for rotary motor.

The vibration characteristics during motor operation were measured using gyroscope 2022 software, as shown in Figure 14. The results demonstrate that the angular velocity deviations are X-axis [27,28]: -0.02 to 0.04 rad/s, Y-axis: -0.04 to 0.04 rad/s, Z-axis: -0.005 to 0.005 rad/s. The minimal angular velocity deviations observed on all axes indicate significantly reduced rotational vibration, confirming stable motor operation and successful dynamic balance optimization of the rotary motor [29].

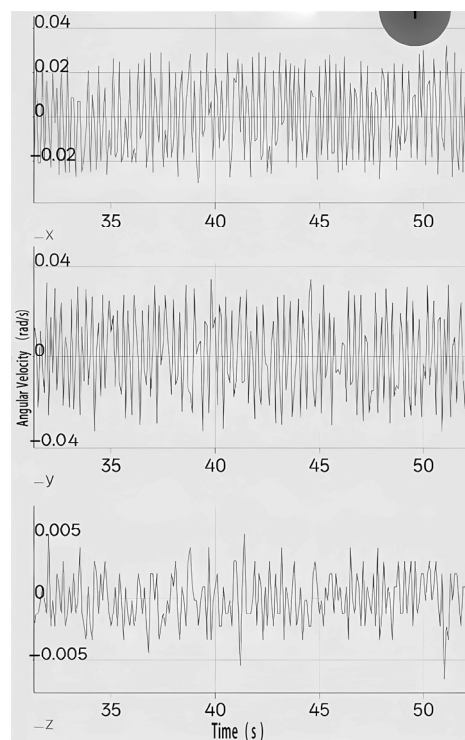


Figure 14. Angular velocity measurement of rotary motor using gyroscope.

5.2. Output Waveform Analysis of Drive Board

The motor developed in this study is a brushless DC (BLDC) motor, with detailed specifications provided in Table 6:

Table 6. The parameters of the brushless DC (BLDC) motor.

Parameters	Values
rated voltage U/V	24
rated current I/A	5
pole pairs p	4
inductance L/mH	6
resistance R/ Ω	3

The hardware circuit was constructed to enable data transmission between the host computer (VOFA+) and the microcontroller unit (MCU, AT32F403) via serial communication. As shown in Figure 15, the SVPWM signals displayed on the VOFA+ platform exhibit a sinusoidal-modulated saddle wave pattern [30,31].

Figure 16 displays the three-phase current values captured by the main control chip and visualized on the VOFA+ host computer, exhibiting a sinusoidal waveform. Experimental results demonstrate the optimized motor achieves significantly smoother rotation.

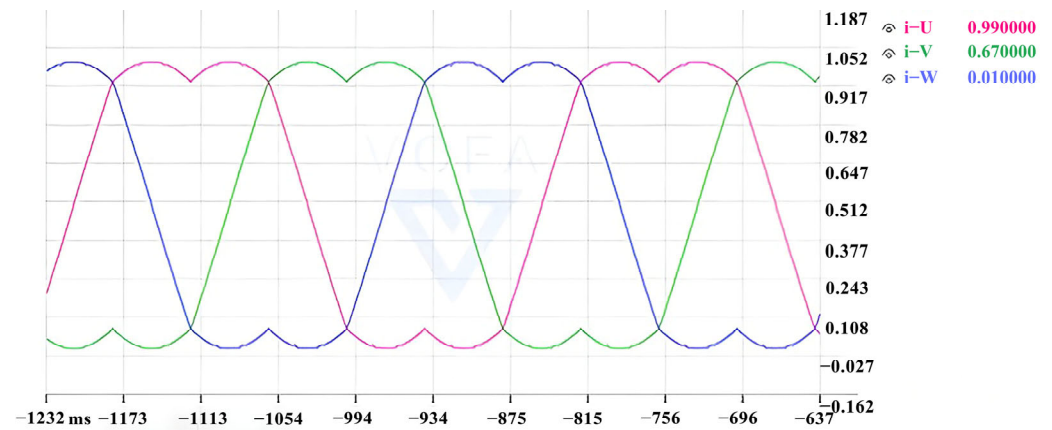


Figure 15. Three-phase saddle-wave SVPWM signals.

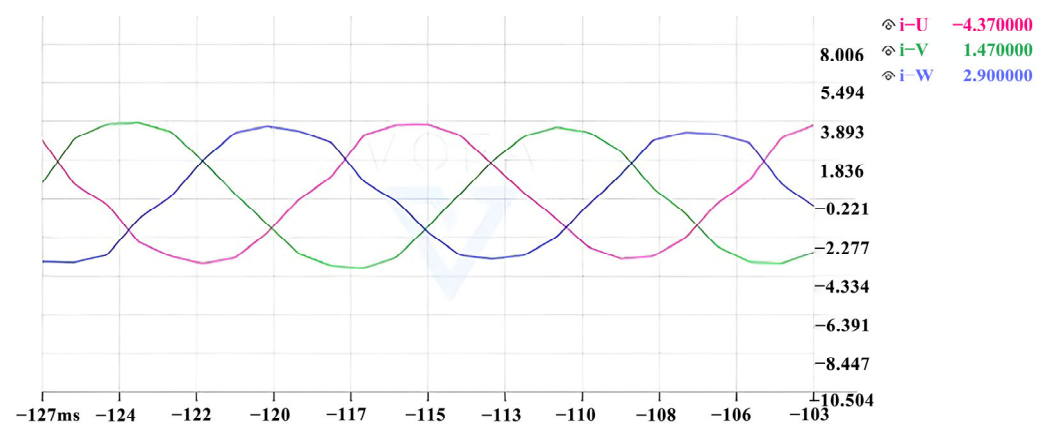


Figure 16. Three-phase current waveforms of rotary motor.

5.3. Plasma Spray Coating Performance Evaluation of Rotary Motor System

As shown in Figure 17, the optimized rotary motor was integrated into the plasma power supply test platform [32]. The dyne pen test performed on the acrylic substrate (Figure 18) demonstrates uniform and stable plasma flame distribution [33,34]. The plasma power supply in the picture is manufactured by Siying Company in Guilin, Guangxi, China, and the air compressor is manufactured by Outus Company in Taizhou, Zhejiang, China.

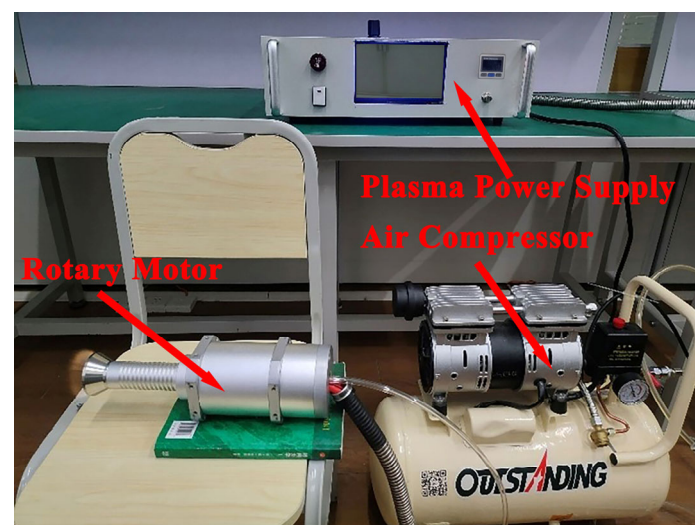


Figure 17. Experimental setup of the rotary motor-integrated rotary motor plasma power platform.

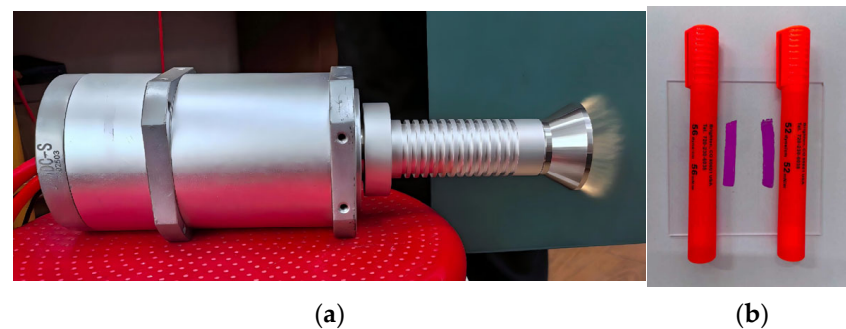


Figure 18. Dyne test results for the motor-assisted plasma spraying process. (a) Plasma flame generated by the motor-driven system; (b) acrylic substrate after plasma spray coating.

Table 7 presents the discrepancies between the simulated and measured values of rotational speed and current. As shown in Table 7, the error between the measured and simulated rotational speed is 0.2%, while the error between the measured and simulated current is 1.7%, both of which meet the design requirements for the rotating electrical machine.

Table 7. The discrepancies between the simulated and measured values.

Testing Metrics	Simulated Values	Measured Values	Error
rotational speed/rpm	2250	2245	0.2%
current/A	3.886	3.818	1.7%

As demonstrated in Table 8, after structural optimization and implementation of the FOC algorithm, the motor-enabled plasma system successfully passed dyne level 52 and 56 tests, achieving satisfactory coating uniformity [35,36]. The effectiveness of plasma spray treatment highly depends on the stability of the interaction between the plasma jet and the workpiece surface. In this study, the combined effects of dynamic balance optimization in the motor structure and high-precision FOC-SVPWM control significantly suppressed vibration of the rotating system and ensured smooth rotational speed. Such improvement in mechanical operating conditions directly translates into a constant distance between the plasma spray gun and the workpiece during spraying, as well as a uniform scanning speed of each point on the workpiece surface through the plasma flame zone. The stability of distance and velocity ensures uniform distribution and consistent action of plasma energy on the workpiece surface, thereby achieving more thorough and uniform surface activation [37,38]. The dyne pen test quantifies surface energy by detecting wettability, and the improvement in its grading provides direct evidence for the systematic enhancement of the aforementioned surface activation effects.

Table 8. Dyne test results of motor-assisted plasma spray coating.

Dyne Pen Test	Pre-Optimization Motor	Post-Optimization Motor
dyne 52	failed	passed
dyne 56	failed	passed

6. Conclusions

This study systematically enhances the performance of the rotational motor in plasma spraying equipment through integrated structural optimization and SVPWM control strategy design. Static simulation using Ansys indicates that the optimized single-bearing support structure reduces the maximum stress in the bearing region from 1.295 MPa to 0.865 MPa, a reduction of 30%, effectively mitigating stress concentration risks and ensuring

long-term operational reliability. Adams dynamic balance simulation further shows that the center-of-mass deviation range of the optimized motor narrows from ± 0.05 mm to ± 0.0175 mm, a 65% reduction, significantly improving rotational smoothness and contributing to vibration and noise suppression. Experimental results reveal an error of 0.2% between measured and simulated rotational speeds, and a 1.7% error between measured and simulated currents, meeting the design requirements for rotating electrical machines.

At the hardware and control levels, the FOC-based SVPWM algorithm was implemented, achieving sinusoidal three-phase current output and smooth motor operation. In experimental tests, the SVPWM signals exhibited well-defined saddle-wave shapes, with phase current waveforms closely approximating ideal sinusoids, ensuring stable motor performance under rated conditions. Further plasma spraying experiments confirmed that the optimized motor system delivered uniform and stable flame morphology, passing dyne 52 and 56 tests, thereby validating its effectiveness and practicality in real spraying tasks.

This study provides practical guidance in the following aspects:

1. Trade-offs between structural optimization and control system complexity: The proposed single-bearing support structure not only reduces weight and cost but also simplifies assembly processes. Compared to traditional dual-bearing configurations, it maintains rigidity and stability while reducing the complexity of subsequent control system tuning. This reflects a design philosophy of “simplifying structure to enhance control optimization,” offering valuable insights for motor design in high dynamic load applications.

2. Methodological scalability: The integrated approach of “stress-dynamic balance co-simulation + SVPWM-FOC control validation” demonstrates strong generalizability and can be extended to the optimization of rotary motor with different sizes and speed ranges. By adjusting simulation boundary conditions and control parameters, the method adapts to diverse engineering needs, highlighting its scalability and application potential.

The study has certain limitations that warrant further exploration in future work:

1. Thermal and fatigue effects under high load or high duty cycle: The continuous heating effects in high-load or high-duty-cycle scenarios, along with their potential impacts on material fatigue and control performance, have not been thoroughly analyzed. Additionally, the influence of bearing stiffness and frequency characteristics on the motor’s dynamic response requires modeling and validation with experimental data.

2. Load assumptions in stress simulation: The current stress simulation primarily considered gravitational and centrifugal loads at rated speed. Future work could incorporate more complex loading conditions, such as dynamic recoil forces during plasma spraying, for transient dynamic analysis to obtain structural stress states closer to actual extreme operating conditions.

3. Systematic testing under dynamic loads: Future studies should include systematic tests on a motor dynamometer platform, simulating step changes in load torque while recording transient responses of motor speed, three-phase currents, and DC bus current using high-precision data acquisition systems. Analyzing these dynamic responses and comparing them with theoretical expectations would more comprehensively validate the proposed control strategy.

Subsequent research will focus on these directions to comprehensively enhance the system’s overall performance and reliability during long-term operation.

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